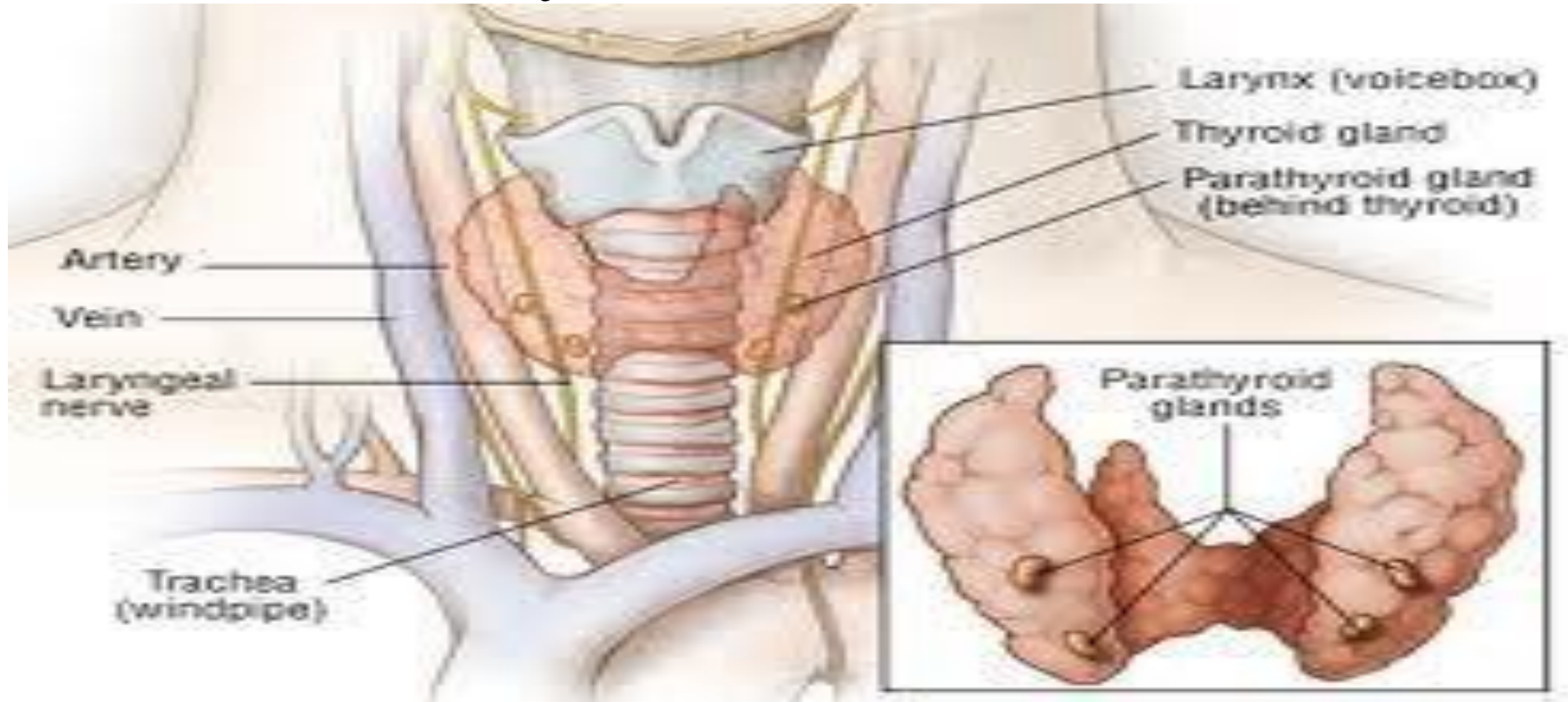


Thyroid Crisis



Prepared by: Assist professor

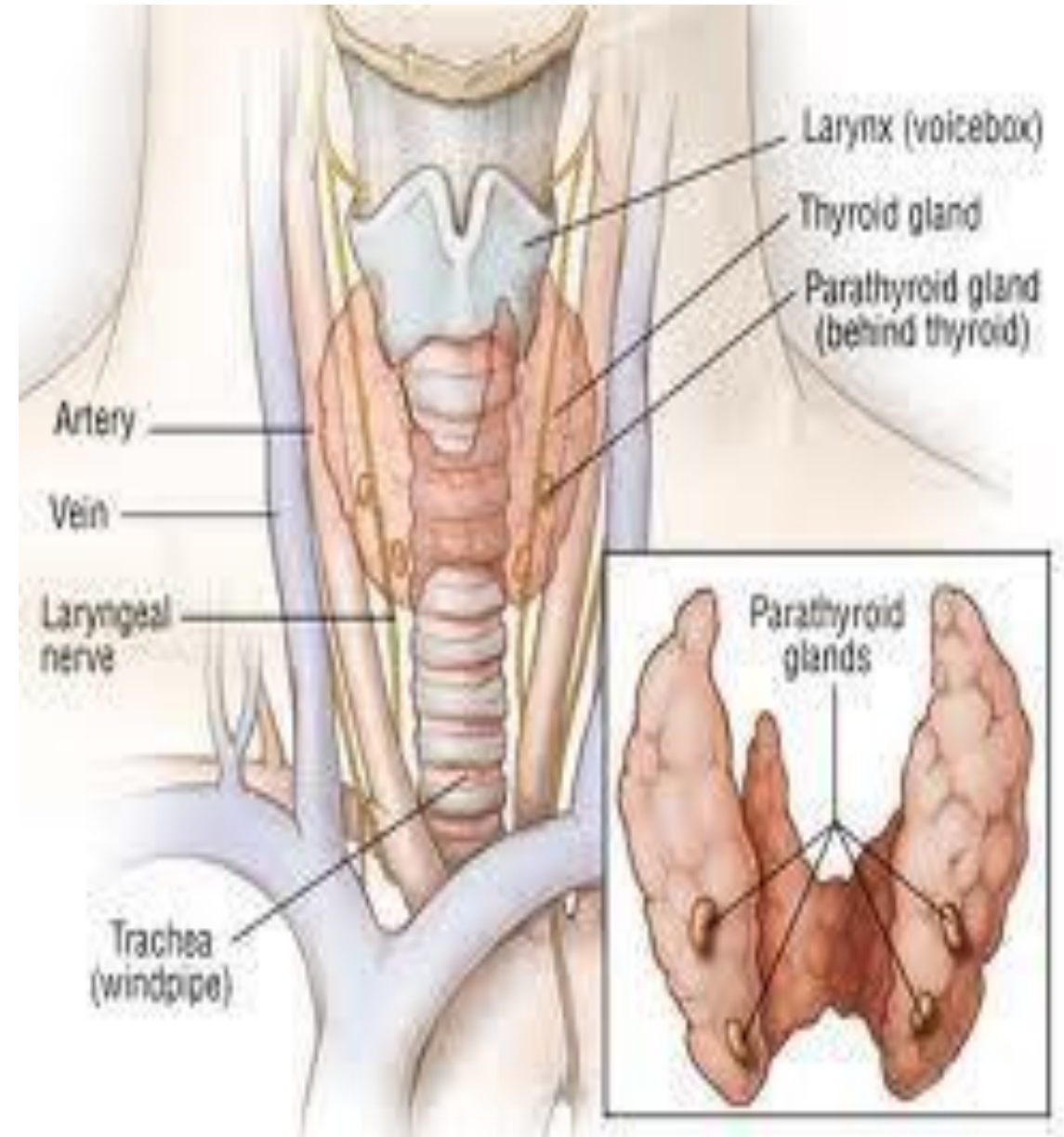
Dr :Afaf Mohamed Mansour

Outlines

- ❖ **Description**
- ❖ **Etiology**
- ❖ **Pathophysiology**
- ❖ **Clinical Manifestations**
- ❖ **Assessment and Diagnosis**
- ❖ **Medical Management**
- ❖ **Pharmacologic Management**
- ❖ **Nursing Intervention**

Introduction

The patients with thyroid storm or crisis is critically ill and requires astute observation , aggressive and supportive nursing care during and after the acute stage of illness. but with proper treatment the mortality rate is reduced substantially.



Thyroid gland

The thyroid is a small, **butterfly-shaped** endocrine gland in the front of the neck, below the **larynx** and **anterior to the trachea**.

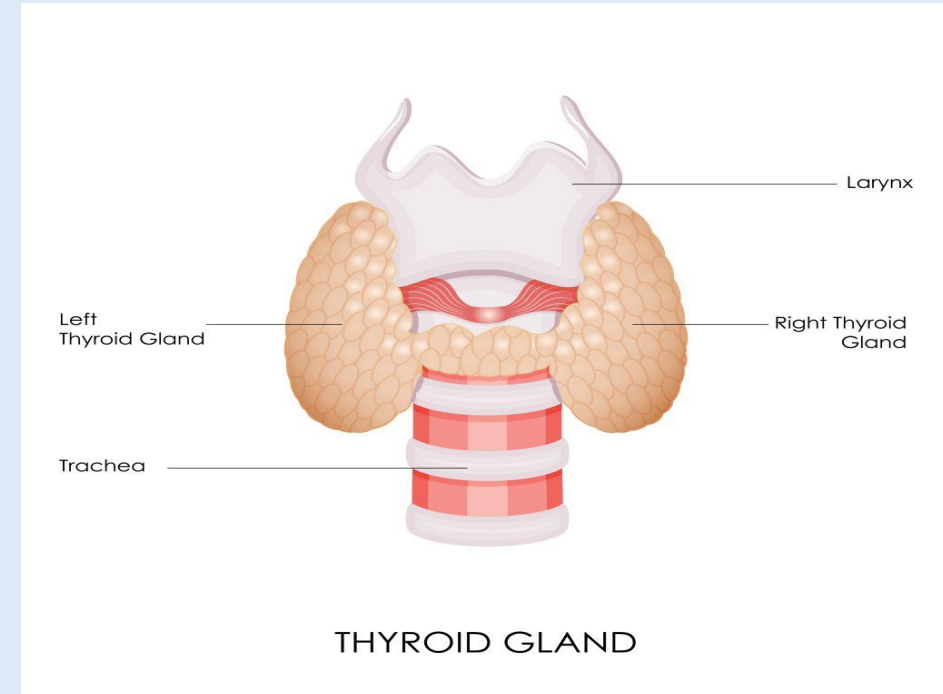
The thyroid gland produces

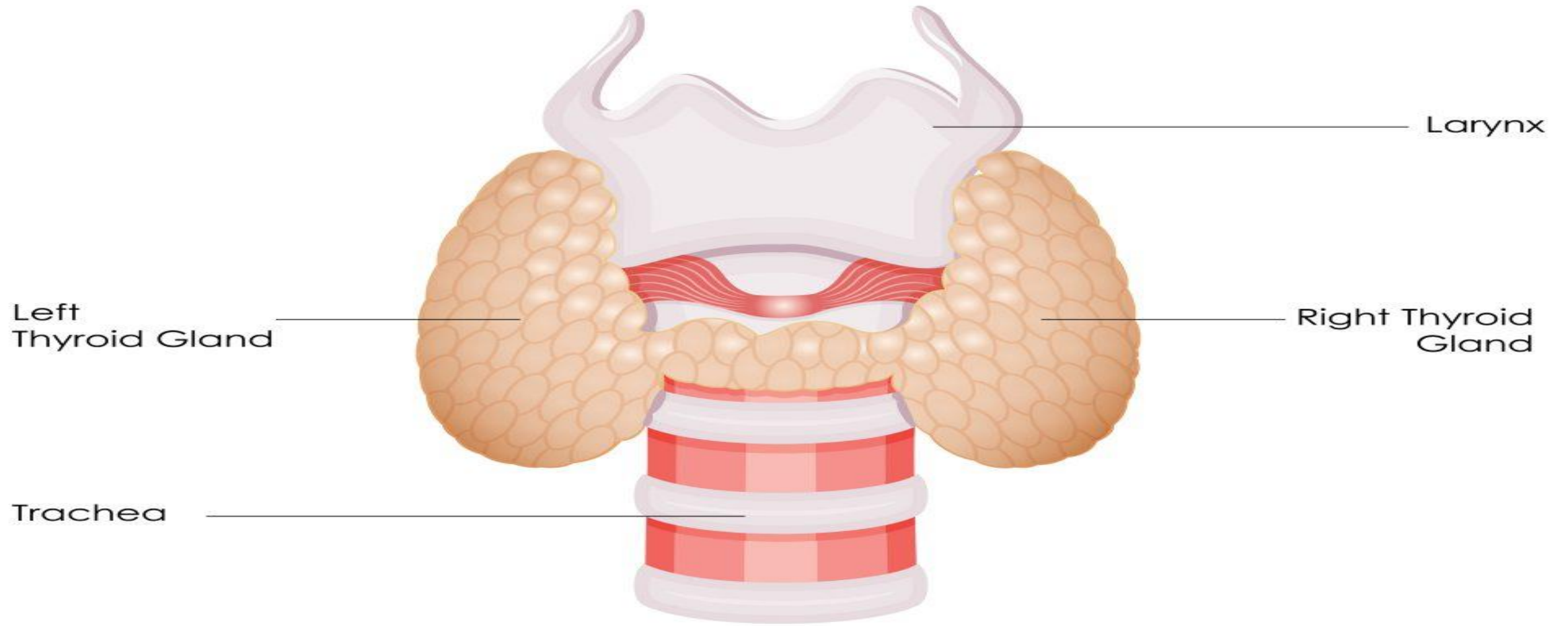
❑ **Two iodine-dependent hormones:**

❖ Thyroxine (T₄)

❖ Triiodothyronine (T₃)

❑ **A third hormone** known as thyrocalcitonin (calcitonin) is produced by the C cells of the thyroid gland in response to calcium levels. The C cell makes calcitonin. that helps to regulate calcium levels in the blood.



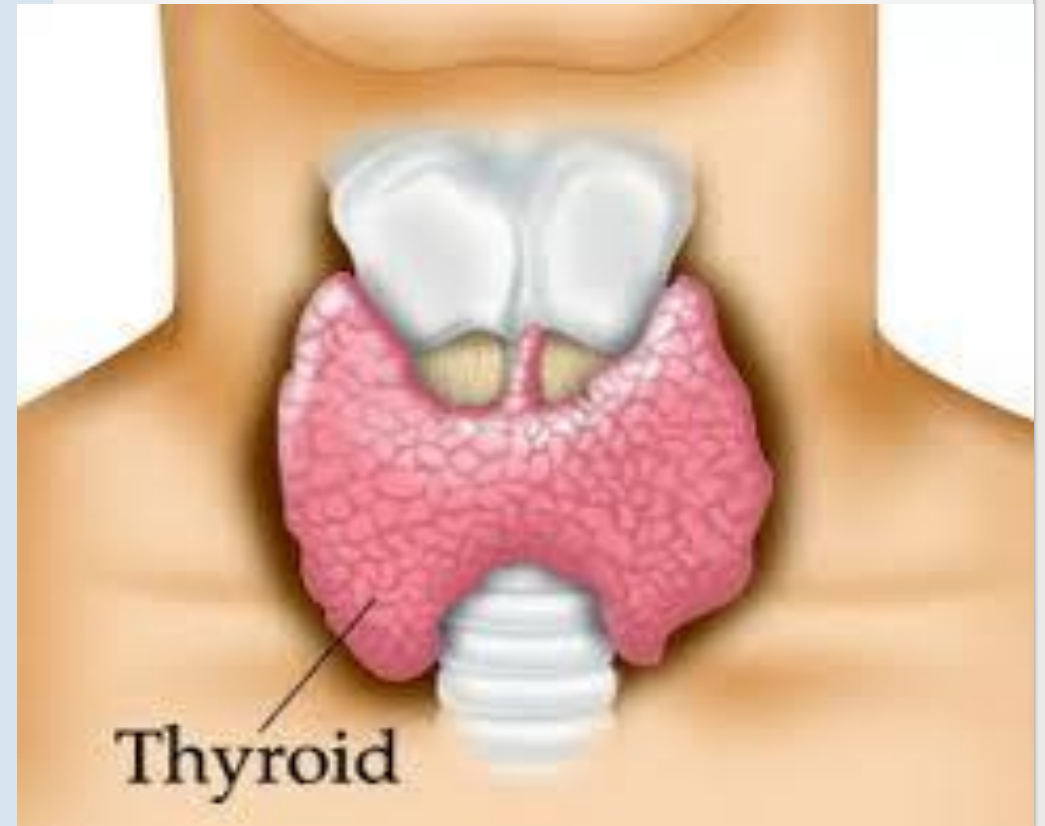


THYROID GLAND

Function of thyroid gland

These hormones play a role in

- Regulating the metabolic processes controlling the rate of growth
- Oxygen consumption
- Contractility of the heart
- Calcium absorption



Thyroid Crisis

- . Thyroid storm (thyrotoxic crisis) is a form of severe hyperthyroidism, usually of abrupt onset. Untreated it is almost always fatal

Etiology

- Over half the causes of primary hyperthyroidism are due to **Graves disease**, an autoimmune condition.
- **Graves disease** is caused by circulating **immunoglobulin-G (IgG) antibodies** that attack the thyroid stimulating hormone (TSH) receptor cells on the gland' causing **release of thyroid hormone**.
- The **antidysrhythmic** medication **amiodarone** causes thyroid dysfunction in 10% to 20% of amiodarone-treated patients.
- **Stopping antithyroid medications**
- **Major stressors**, such as **infection, surgery, trauma, pregnancy, or critical illness**, can precipitate thyroid storm in any patient with pre-existing hyperthyroidism.

Etiology

- ❑ In thyroid storm, **excessive endogenous thyroid hormone**
 - Increases metabolic activity
 - Stimulates the beta-adrenergic receptors resulting in a heightened SNS response. There is hyperactivity of cardiac tissue, nervous tissue, and smooth muscle tissue and tremendous heat production

Pathophysiology

Precipitating Factor



An Increased Cellular Response To Thyroid Hormone



Hyperactivity Of The Sympathetic Nervous System



Increased Response To Catecholamine



Intense Metabolic Activity That
Increases Oxygen Requirements



Causing Cytokines Release And
Altered Immunological
Disturbances



Multi Organ Dysfunction



Hyperbension
Tachycardia
Hyperthermia
Increased BMR
Sweating
Heart Failure



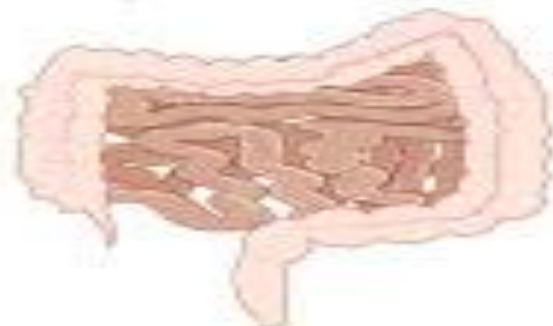
Dilation of
bronchioles



Increased
heart rate

CATECHOLAMINES

Decreased digestive
system activity



Liver converts
glycogen to glucose

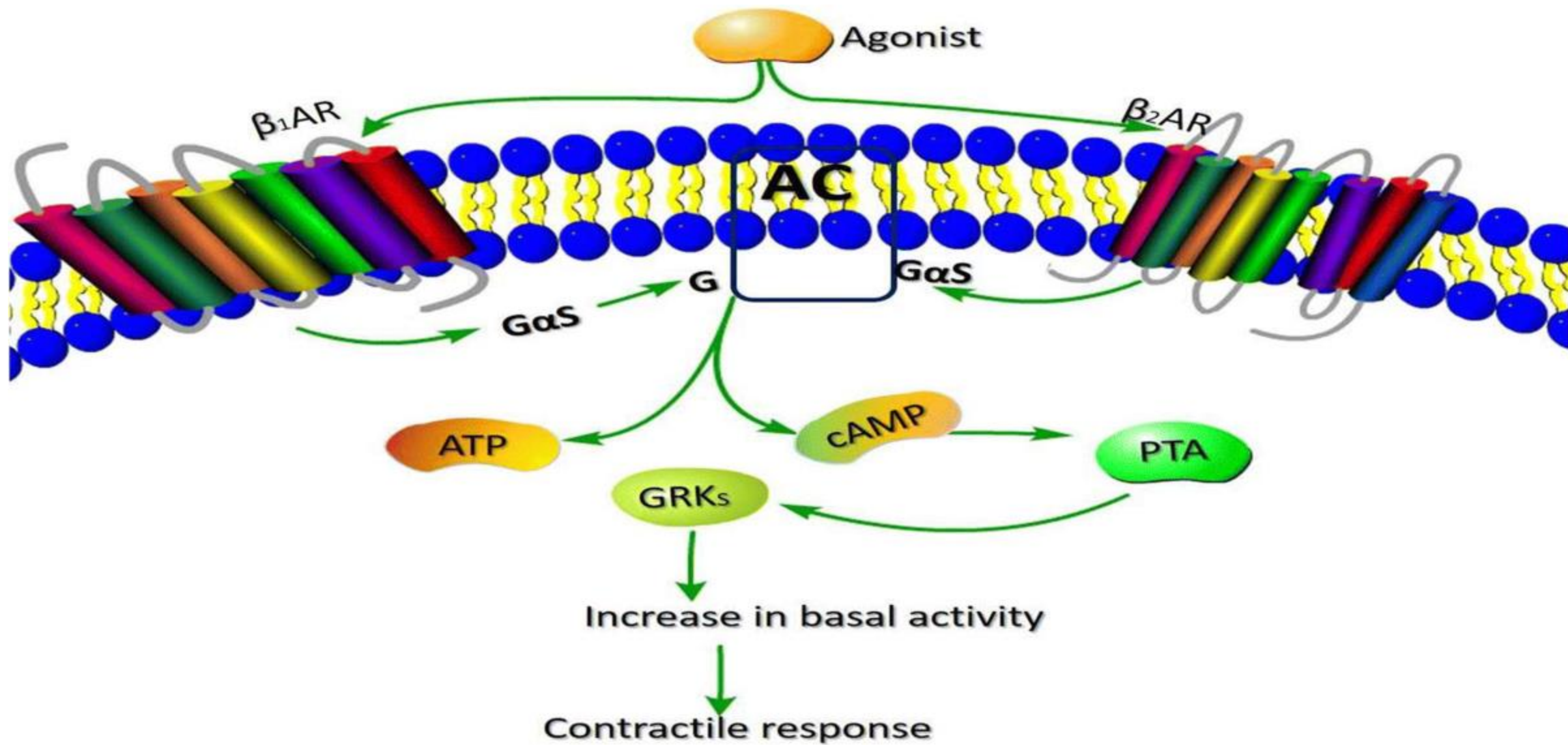


High blood pressure

Clinical Manifestations

I. **Cardiovascular System Prompted by Increased Affinity of Beta-Adrenergic Receptors in the Heart.**

- Tachycardia
 - Increase stroke volume
 - Increased systolic blood pressure
 - Extra systoles
 - Premature ventricular contraction
 - Chest pain
 - Acute heart failure
 - Cardiogenic shock
- systolic murmur .
- Increased cardiac output
- Decreased diastolic blood pressure
- Paroxysmal atrial tachycardia
- Palpitations
- Increased cardiac contractility
- Pulmonary edema.

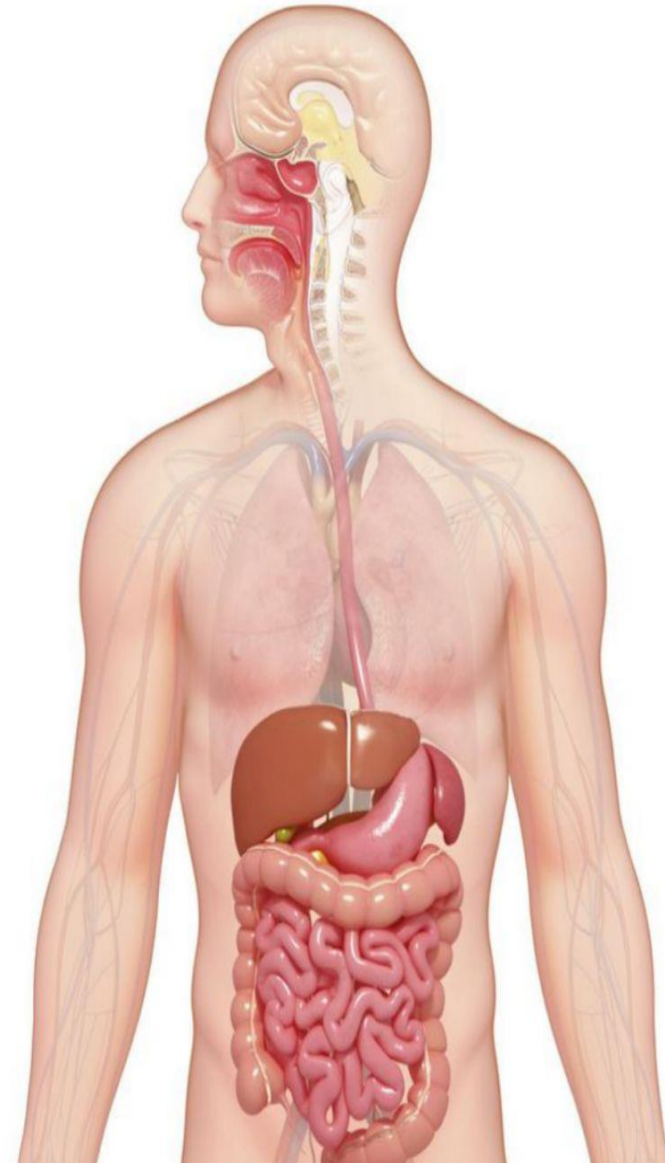


2- Central Nervous System Resulting from an Increased Catecholamine Response

- Hyperkinesia
 - Muscle weakness
 - Convulsions
 - Fine tremor
 - Frank psychosis
- Diaphoresis
- Nervousness
 - Confusion
 - Heat intolerance
 - Emotional lability
 - Apathy

Gastrointestinal System

- Nausea .
- Vomiting
- Diarrhea
- Liver enlargement
- Abdominal pain
- Weight loss

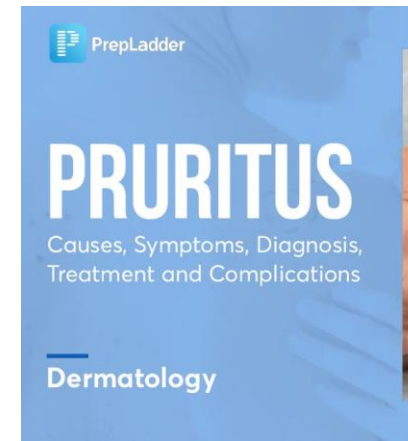


I. integumentary System

- Pruritus
- Hyperpigmentation of skin
- Fine, straight hair
- Alopecia.

V-Thermoregulatory System

- Hyperthermia
- Heat dissipation
- Diaphoresis



Alopecia.

Serum or Urine

- Hypercalcemia
- Hyperglycemia
- Hypoalbuminemia
- Hypoprothrombinemia
- Hypocholesterolemia
- Creatinuria



Assessment and Diagnosis

The clinical presentation of thyroid storm is characterized by a variety of organ systems including:

- 1- **Thermoregulation: fever**
2. **Heart:** atrial fibrillation, supraventricular tachycardia, acute heart failure
3. **Central nervous system:** agitation, restlessness, delirium
4. **Gastrointestinal:** nausea, vomiting, diarrhea, unexplained jaundice, stupor, coma

Symptoms of a Thyroid Storm



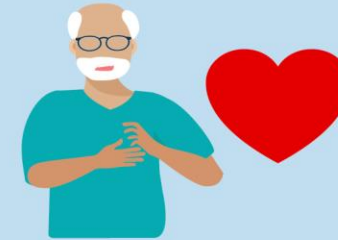
High fever.



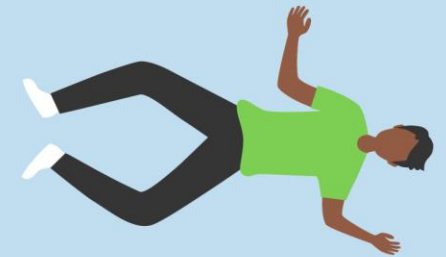
Agitation.



Delirium.



Congestive heart failure.



Loss of consciousness.



Seek immediate medical care if you're experiencing these symptoms.

Medical Management

- The goal of acute medical management of thyroid storm is
 - ✓ To reduce the clinical effects of thyroid hormone as rapidly as possible.
 - ✓ Preventing cardiac decompensation,
 - ✓ Reducing hyperthermia,
 - ✓ Reversing dehydration caused by fever or gastrointestinal losses.

Medical Management

Prevent Cardiovascular Collapse

- Atrial dysrhythmias need to be controlled and progression of heart failure halted. **Beta-blockers** are the mainstay of therapy for cardiac protection.

Reduce Hyperthermia

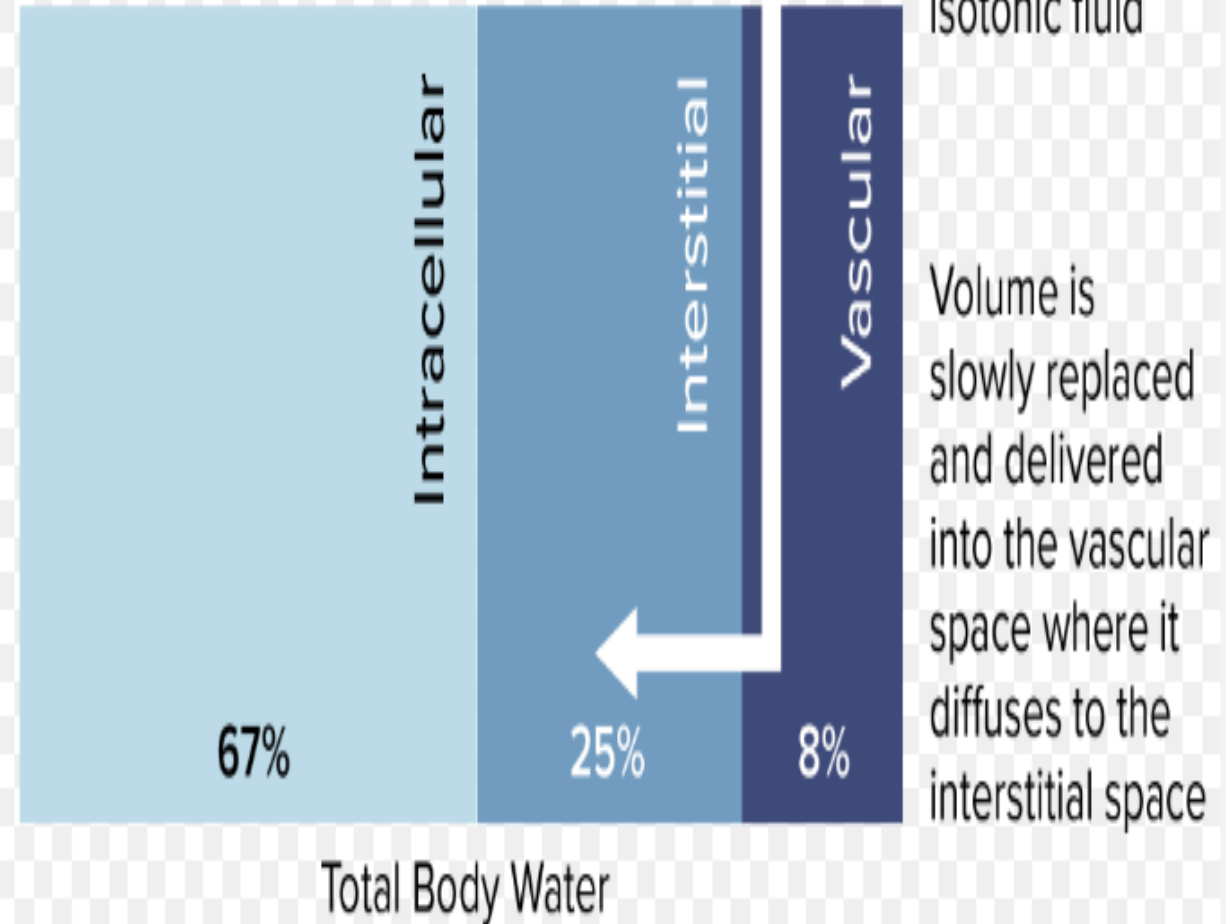
- Use of a cooling blanket
- Antipyretic agent acetaminophen.
- **Salicylates (aspirin)** are contraindicated because they inhibit protein binding of T3 to T4, increasing the level of free, metabolically active, thyroid hormone.

Medical Management

Reverse Dehydration

- Dehydration is treated with large volumes of glucose and isotonic sodium solutions to replace circulating fluid and sodium losses from hypermetabolism.
- Antibiotic therapy may be warranted in the presence of systemic infection.

Treatment of Dehydration



Pharmacologic Management

- **Beta-blockers** are administered to decrease the peripheral cellular sensitivity to catecholamines
- **Antithyroid medications** are administered to block the synthesis and release of thyroid hormone into the circulation

Pharmacologic Management

1- Block the Catecholamine Effect

To decrease the catecholamine effects of excessive thyroid hormone, beta-adrenergic blocking agents are used.

- **Propranolol** is the most frequently used medication
- Beta-blockers** have no effect on thyroid hormone but reduce the exaggerated myocardial stimulation and slow the atrioventricular (AV) conduction rate.

Pharmacologic Management

2-drugs Block Thyroid Synthesis

*The synthesis of new thyroid hormone is blocked by the administration of **antithyroid medications**.*

- Two agents are frequently used: **Propylthiouracil (PTU)** and **Methimazole**.
- These medications are administered by mouth or via a feeding tube PTU is especially therapeutic because it blocks thyroid synthesis
- Both medications act **within 1 to 2 hours after absorption** from the gastrointestinal tract.

Pharmacologic Management

3-Block Release of Thyroid Hormone

- Administration of **inorganic iodine blocks** the release of any preformed thyroxine that is already in the thyroid gland but not yet released.
- It is essential that iodine therapy not be administered until adequate inhibition of new hormone synthesis has occurred
- The iodide preparations are rapid acting and have a short duration. They are given **approximately 1 hour after administration** of the antithyroid medications.

Pharmacologic Management

Some patients with thyroid storm have concomitant **adrenal insufficiency** and may be prescribed **hydrocortisone** or **dexamethasone** during the initial stages of thyroid storm management.

Drug	Dose	Action	Nursing Considerations
Propylthiouracil (PTU)	Loading: 800–1,200 mg Maintenance: 100–400 mg every 4–6 h orally (PO)	Blocks synthesis of hormones (conversion of T_3 to T_4)	Monitor cardiac parameters. Observe for conversion to hypothyroidism. Must be given by mouth. Watch for rash, nausea, vomiting, agranulocytosis, lupus syndrome, hepatitis.
Methimazole	10–20 mg every 6–8 h PO	Blocks synthesis of thyroid hormone	More toxic than PTU. Watch for rash and other symptoms as for PTU.
Sodium iodide	1 g every 12 h (intravenous [IV])	Suppresses release of thyroid hormone	Given 1 h after PTU or methimazole. Watch for edema, hemorrhage, gastrointestinal upset.
Potassium iodide	2–5 gtts every 8 h PO	Suppresses release of thyroid hormone	Discontinue for rash. Watch for signs of toxic iodism.
SSKI	5–10 gtts every 8 h PO	Suppresses release of thyroid hormone	Mix with juice or milk. Give by straw to prevent staining of teeth.
Dexamethasone	2 mg every 6 h IV	Suppresses thyroid hormone release	Monitor intake and output. Monitor glucose. May cause hypertension, nausea, vomiting, anorexia, infection.
β -Blocker (eg, propranolol)	1–3 mg every 1–4 h IV	β -Adrenergic blocking agent	Monitor cardiac status. Hold for bradycardia or decreased cardiac output.

Nursing interventions;

Nursing interventions are directed toward

- ❑ Safe administration and monitoring of the effects of prescribed medications
- ❑ Normalizing body temperature
- ❑ Rehydration with correction of other metabolic derangements and provision of patient education.

Medication Administration

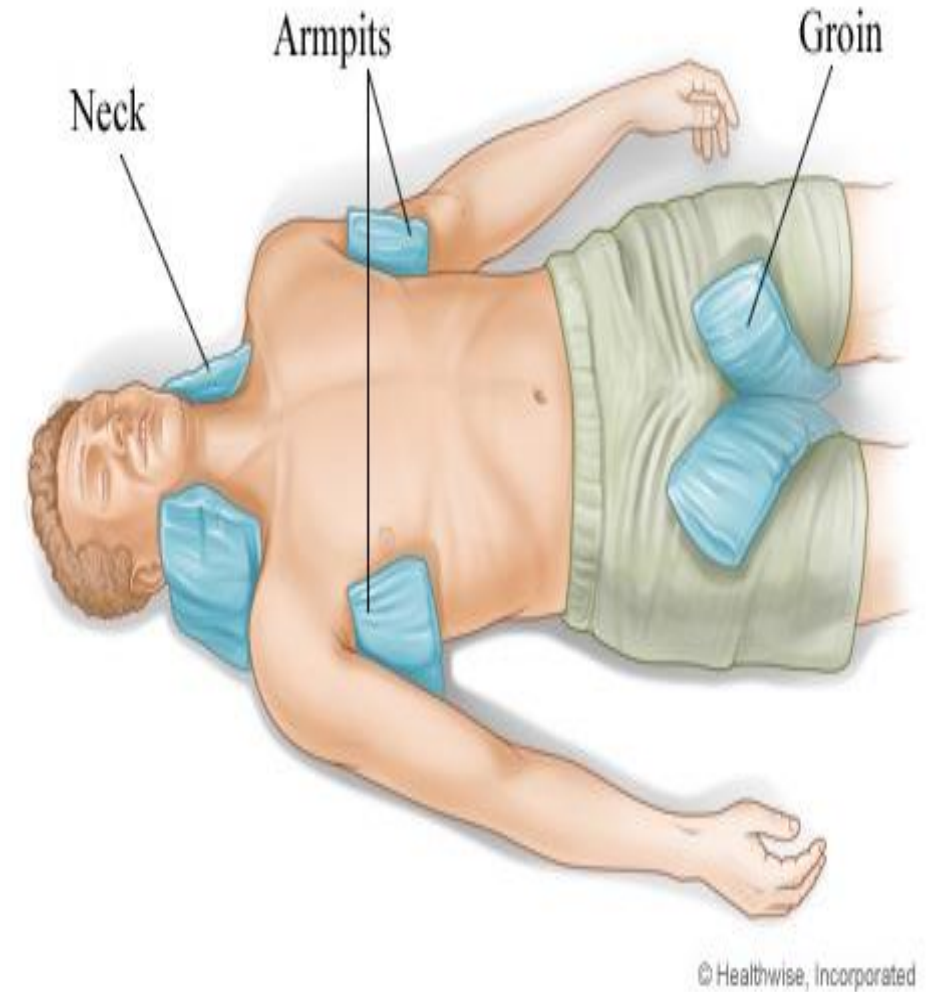
- The timely and ordered sequence of medication administration is essential in the management of thyroid storm
- The patient in thyroid storm is **agitated, anxious, and unable to rest** and benefits from an **environment that is calm**.
- Heart rate should decrease with beta-blockade.
- The patient and family need to be **reassured** that this extreme agitation is the result of the disease process and that the medications will help control the nonstop fidgeting and tremors.
- **Frequent reassurance and clear, simple explanations** of the patient's condition help decrease the fear brought on by the onset of thyroid storm

Normalize Body Temperature

- In thyroid storm, the patient has hyperthermia related to a hypermetabolic state, as evidenced by a critically high body temperature; diaphoresis; hot, flushed skin; intolerance to heat tachycardia; and tachypnea.
- Temperature is assessed frequently until normal body temperature is attained.
- Nursing measures to provide comfort while the patient is intolerant to heat include **a room with a cool environment** and **a fan** to circulate air, and comfortable, **nonrestrictive bedclothes**.

Normalize Body Temperature

- A tepid sponge bath helps to reduce heat by evaporation, and cold-pack applications to the groin and axilla increase heat loss at major blood vessels.
- If antipyretic medications are required, acetaminophen is the agent of choice



Rehydration and Correction of Metabolic Derangements

- Hyperthermia, tachypnea, diaphoresis, vomiting, and diarrhea predispose the patient to **a fluid volume deficit**. Fluids and electrolytes are as vigorously replaced as the decompensated cardiovascular system can tolerate.
- Glucose solutions are given to replace glycogen stores, which are depleted. **Insulin is administered** to treat the hyperglycemia that results from mobilization of nutrients and glucocorticoids.
- Hyponatremia from active loss (e.g., vomiting) is monitored by means of laboratory serum values. Hyponatremia can be prevented or treated with **isotonic IV fluid** replacement

Patient Education

- During the critical events surrounding the thyroid storm, the patient and family are given information according to their emotional state and cognitive level of understanding.
- **The cause of the high fever, anxiety, and cardiac dysrhythmias** is explained in understandable terms
- Side effects of the specific medication therapy are **taught** before discharge

Patient Education

- Patients treated with beta-blockers are **taught to report signs'** of bradycardia, unexplained fatigue, and orthostatic hypotension, among other untoward effects.
- Patients discharged with antithyroid medications are alerted to the potential side effect, **agranulocytosis**.
- Symptoms of agranulocytosis include **sudden cough, fever, rash, and inflammation**.
- For management of elevated temperature, patients are instructed to use **acetaminophen rather than salicylates**, because salicylates increase the amount of free thyroid hormone in circulation.

MCQ

Which of the following findings would most likely indicate the client is experiencing a thyroid storm?

- A. Bradycardia and hypotension
- B. Hyperpyrexia and extreme tachycardia
- C. Numbness and tingling in the fingers
- D. Lethargy and cold intolerance

What is the expected outcome of an untreated thyroid storm?

- A. Spontaneous recovery
- B. Progression to mild hyperthyroidism
- C. It is almost always fatal
- D. It converts to hypothyroidism

